

Term paper

Assignment in

Business Simulation

Analysis

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GRA4138 Business Simulation Analysis

Programme:

MSc in Business Analytics

PART A: Monte Carlo Simulation for Vineyard Investment

Summary:

In This report we evaluate five vineyard investment alternatives combining grape varieties and farming techniques. Using Monte Carlo simulation over a 20-year horizon, we estimated each option's Net Present Value (NPV), accounting for yield, demand, prices, costs, and disease risk. We recommend Alternative 5 (Variety A & C, Modern farming), which provides the highest mean NPV of 3,15 billion CLP and the lowest downside risk, making it suitable for both risk-neutral and risk-averse investors.

A) Risk-Neutral Analysis:

The best investment alternative for a risk-neutral family business aiming to maximize profitability is alternative 5. This option is a 50/50 mix of Variety A (Carménère) and Variety C (Syrah), using modern farming techniques across the full 20 hectares. Based on the results from 1000 Monte Carlo simulations, alternative 5 achieved the highest average NPV of approximately 3,14 billion CLP over the 20-year planning horizon. It consistently outperformed all other combinations, including those based on traditional farming methods or single-variety planting. Since the family is risk-neutral and focused purely on maximizing expected financial return, this alternative is the most attractive one to go for. It not only offers the highest average profitability, but also shows a potential high maximum performance, with a NPV of approximately 3,63 billion CLP. Additionally, the alternative had the lowest standard deviation of 149572815,6 and 95% confidence interval range of 18 540 897,62 CLP explaining the accuracy of this option. For a business focused on long-term financial gain, Alternative 5 delivers the best expected return and is therefore the recommended alternative to choose.

B) Risk-Averse Analysis:

If the family owners are risk-averse and wish to minimize the chance of receiving a low NPV, we must evaluate how likely each alternative is to produce a Net Present Value below 1,7 billion CLP. After running 1000 simulations for each alternative, the analysis showed that none of the five alternatives had a minimum NPV below this threshold. This means that even in the worst-case simulated

scenario, all alternatives are expected to deliver a positive financial return higher than 1,7 billion CLP. To evaluate the minimum value, we calculated the probability of achieving an NPV score lower than 1,7 billion CLP, and the result for all alternatives was 0%. We compare the actual NPV with the average NPV over the 1000 samples and can see that it's not deviating much, this is a good indicator that the value is good. Also, alternative 5 had the lowest difference, which again explains how well this alternative is. Therefore, Alternative 5 also suits risk-averse investors and is the most appropriate choice. It produced the highest minimum NPV, around 2,62 billion CLP, and offered the strongest protection against poor performance, with a low standard deviation of 149.572.815,6 CLP.

Model explanation:

The model simulates how different vineyard investment alternatives might perform over a 20-year period. It combines randomly generated values in grape yields, prices, costs, and disease events using realistic probability distributions. For each of the five alternatives, it calculates annual profits and then uses a discount rate of 6% to estimate the total NPV. This process is repeated 1,000 times per alternative to reflect uncertainty and provide a reliable prediction of future profits. Alternative 5, which combines Variety A and C using modern farming techniques, shows a strong and stable performance. We compared the different NPV values for each alternative and interpreted the mean, confidence level, std, and minimum and maximum values. The histograms (figure 1-5) show average NPV normally distributed with most outcomes concentrated between 3,12 and 3,18 B CLP for all alternatives. This suggests a relatively low spread and normal variation, indicating predictability and consistent returns of NPV for all alternatives.

To simulate realistic outcomes, the model generates random values for input parameters based on appropriate probability distributions. Yields are modeled using triangular distributions specific to each grape variety. Prices follow a normal distribution with defined means and standard deviations. Material costs use a lognormal distribution. Annual demand is based on a normally distributed growth factor applied to base demand levels. Disease events are included with a 5% annual probability, and yield reductions from disease are sampled from an

exponential distribution depending on the farming technique used. Profit each year is calculated as the minimum of adjusted yield and demand, multiplied by the price per ton and total area (20 hectares), minus fixed and variable costs. The resulting cash flows are then used to calculate NPV using Excel's NPV function.

Appendix: Summary statistics & Histograms

Descriptive statistics:

Mean NPV we compare with actual NPV. Its not a very big difference, wich means that the 1000 samples is not deviating to much. Change between actual NPV and average NPV: 181 629 371 CLP. (if its pretty close to actual value its a good indactor for the model working well).

Standard deviation: The lower std, the less risky is this project.

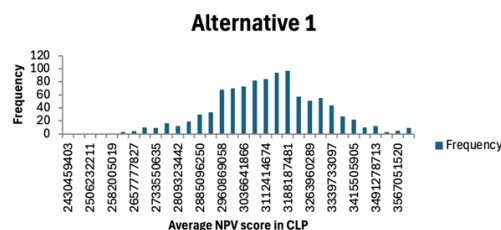
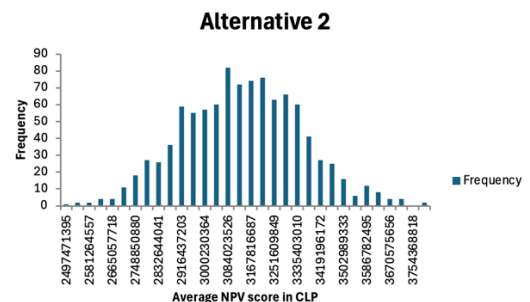
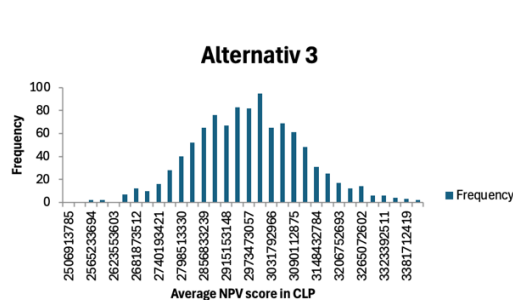
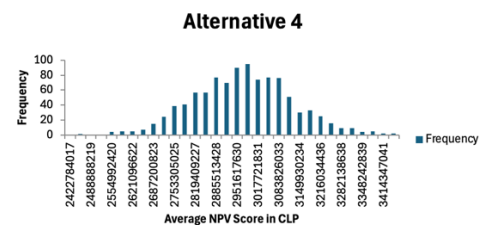
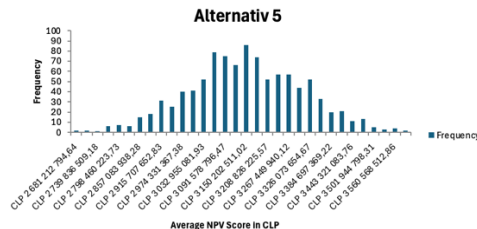
Kurtosis: Should be in the range -2 to 2. Show the validity of the project.

Skewness: Should be in the range of -1 to 1. That means it is normally distributed.

We have a positive minimum NPV wich shows us that all our NPVs are postive! So it means that in a worst casue scenario our lowest NPV for this spesifci alterative of chosen grapes and method is 2 521 456 634 CLP. The best possible NPV that can be reached is 3 674 383 220.

Main Statistics	Alt 1	Alt 2	Alt 3	Alt 4	Alt 5
Mean NPV	CLP 3 093 024 911,87	CLP 3 119 167 886,13	CLP 2 957 658 048,56	CLP 2 960 814 212,91	CLP 3 143 510 833,39
95 % CL range	CLP 22 467 362,28	CLP 27 692 376,87	CLP 16 153 996,01	CLP 20 032 438,83	CLP 18 540 897,62
Standard Deviation	CLP 181 248 324,93	CLP 223 399 474,26	CLP 130 317 243,37	CLP 161 605 351,63	CLP 149 572 815,63
Min	CLP 2 521 456 534,09	CLP 2 226 369 532,64	CLP 2 559 314 875,43	CLP 2 420 457 532,37	CLP 2 624 939 572,34
Max	CLP 3 674 383 220,75	CLP 3 790 039 411,07	CLP 3 372 670 663,07	CLP 3 467 021 160,06	CLP 3 635 279 698,55
Probability of NPV < 1,7 Billion	0,00 %	0,00 %	0 %	0,00 %	0,00 %
Count	1000	1000	1000	1000	1000

Histograms of average NPV values for all alternatives 1-5:



Part B: LactoTrans Distribution Network Optimization

Non-technical section:

We have analyzed LactoTrans current status by doing a blueprint of their logistic operations, even though there are some assumptions made to reach the final goal, our approach was aimed to be realistic to collect valuable and useful insights from this data.

The simulation of the new warehouse reveals that it will likely make the Iasi warehouse obsolete, as trucks will no longer service that location. Additionally, this change is expected to increase overall expenses. During the simulations, we also increased the maximum demand of two warehouses at random to test the system's capacity. Despite this increase in demand, costs rose significantly, confirming that operating the new warehouse under current conditions is not cost effective. Therefore, we do not recommend opening the new warehouse unless there is a significant expansion in the number of retailers served.

A major issue identified is excessive product spoilage, which demands immediate attention to prevent significant losses in perishable goods. Furthermore, the current truck fleet size is insufficient for the number of retailers being served. If a new factory is opened, increasing the number of trucks will be necessary. At present, late deliveries are causing customer dissatisfaction, indicating a critical need to improve delivery punctuality.

Efficient inventory management and order logistics are fundamental pillars for the profitability and competitiveness of distribution companies. The simulation model developed to control orders and deliveries of perishable dairy products highlights several critical areas for improvement.

One of the key findings is that linking order placement directly and immediately to delivery receipt reduces operational flexibility and can generate inefficiencies. Always receiving deliveries for distributors at maximum capacity can lead to unnecessary logistics and storage costs, such as increased spoilage and higher inventory holding expenses. Managing the delivery receipt process separately, by considering fixed time windows for receipt, actual storage capacity, current

demand, and supply chain constraints, allows for strategic scheduling of deliveries. This approach would avoid overloads and optimizes cash flow by minimizing overordering, and it's also more realistic than ordering the maximum capacity every single week as we have to consider the financial position of the company to afford this event.

Transforming static inventory parameters into dynamic variables was another important step. This ensures that replenishment decisions are based on real, up-to-date stock data that reflects actual consumption and deliveries. Consequently, this reduces both stockouts and excessive stock accumulation. Accurately recording the fixed and variable costs associated with each order, including the value of the product and order processing expenses, is vital for evaluating the economic impact of logistics decisions. This information supports the design of order policies that balance order frequency and volume to minimize overall costs.

LactoTrans faces common challenges related to truck fleet management and delivery route planning. Based on the simulation, trucks should be allocated dynamically to warehouses according to actual demand instead of fixed numbers. This flexible allocation allows the company to adjust the fleet monthly, optimizing fixed costs. For delivery routes, a nearest neighbor strategy is recommended, where each truck starts deliveries at the closest retailer with pending orders and continues to the next closest until all deliveries are complete. This simple logic reduces total travel distance and variable costs such as fuel and vehicle wear.

The model also supports multiple delivery tours per truck within a workday, respecting working hours and labor regulations. If a delivery is in progress when work hours end, the truck completes it but pauses the remaining deliveries until the next business day, avoiding overtime expenses while maintaining service continuity. Loading and unloading times, realistically modeled between 20-45 minutes and 15-30 minutes respectively, significantly impact efficiency. Improving these processes could increase daily operational capacity without requiring additional trucks.

Tracking the total distance traveled by each truck during its tours enables better management of variable costs and identification of inefficiencies. This data is

crucial for informed fleet management decisions and justifies potential investments in route optimization technologies or staff training.

Implementing separate management for orders and deliveries, using dynamic inventory variables, carefully considering all related costs, and maintaining continuous monitoring will allow LactoTrans to significantly enhance logistics efficiency and profitability. Dynamic fleet allocation and strategic route planning will reduce costs and improve customer service. Ongoing analysis and adjustment ensure that the company remains adaptable to market fluctuations and competitive within the distribution sector.

In conclusion, we can say that there are multiple suggestions of different types for the company. Regarding the simulations proposed at the beginning of this work, we consider them feasible only if the number of retailers is increased, considering the costs that this new opening and increased capacity would entail. Problems of spoilage, delayed orders, and the rest of the variables described above must be considered to improve our customer satisfaction and the company's performance. Finally, the truck fleet could operate with a simpler system to prevent warehouses from meeting greater demand than others or trucks from traveling unnecessary distances just to meet a work metric.

Technical section:

MODEL CONCEPTUALIZATION

The development of the simulation model began by creating and refining the internal logic of the Retailer and Distributor agents to accurately reflect the dynamics of demand and supply in a logistics chain. In the Retailer agent, key parameters were defined that represent average daily demand for specific products such as milk, yogurt, and cheese. These parameters were initially linked to an external data table containing information segmented by location, ensuring that each Retailer had realistic and differentiated values based on its region. This integration was achieved by directly assigning values from the table to the parameters declared in each agent instance.

The critical component of the Retailer agent was the `generateDailyDemand()` function, programmed to generate random daily demand for products based on

normal distributions. In technical terms, the standard deviation for each product was calculated as 15% of its mean to reflect natural variability in consumption. The simulation environment's own `normal()` function was used to obtain random values with the specified mean and standard deviation. However, a common problem was identified: the generation of negative values, given that the normal distribution is symmetric and allows results less than zero, which makes no sense for physical demands. In a first attempt, `Math.max(0, value)` was used to truncate these negative values to zero, but this generated a high frequency of zeros in the demand, affecting the model's plausibility. To solve this problem, a helper function was designed and coded.

Additionally, logic was incorporated to handle variations in weekly demand. The current day was calculated within the model using the `time()` function divided by the number of seconds in a day and assigning numerical codes from 0 to 6 to represent the days Sunday through Saturday. For Saturday (day 6), in particular, the `generateDailyDemand()` function generates additional demand for Sunday, adding both values to form the total order for the day. This results in the retailer placing larger orders to cover the weekend, a common practice in real world distribution. To achieve this, local variables were generated to store this additional demand and added to the total order. These values were ultimately assigned to internal variables such as `pendingMilkOrder`, `pendingYogurtOrder`, and `pendingCheeseOrder`, which represent the pending orders for the Distributor.

In the Distributor agent, the logic was implemented to receive and process orders from the Retailer population. Correct communication between agents was ensured, verifying that orders stored in the Retailers were accessible and updatable by the Distributors. This involved establishing a clear protocol for shared or accessible variables and ensuring time synchronization so that orders reflected the actual demand generated at each time step in the model.

During development, traceability messages (`traceln`) were used extensively to print the evolution of critical values such as averages, generated demand, days of the week, and orders to the console, we can visualize this in “*figure 5*” of the appendix. These debugging messages allowed us to detect inconsistencies, such as zero demand values for products with high averages, which led to the implementation of

the function to avoid these unjustified zeros. We also validated that the calculations for the day of the week and the sum of Saturday demand worked correctly.

The parameters, which are initialized with fixed values from the data table and represent static information for the Retailer, were clearly differentiated from the internal variables that are dynamically modified throughout the simulation, such as generated daily demand and pending orders. This separation allowed for orderly management of the agent's status and facilitated the extension of the model to include new logic or products without disrupting the data structure.

In summary for both agents, the technical work performed included reading parameters from external tables to configure agents, controlled generation of random variables following a truncated normal distribution to avoid nonphysical values, the implementation of logic based on days of the week to simulate realistic behavior in weekend orders, and the robust integration of communication between retailers and distributors through shared variables and synchronization. All of this was supported by debugging and testing techniques to ensure data consistency and that the simulation responded to the expected logic, providing a solid foundation for future analysis and optimization of the logistics system.

On the trucks agent we focused on modeling and simulating the logistics operation of LactoTrans's truck fleet as seen in “*figure 4*” in the appendix, aiming on vehicle allocation, route planning, and operational management to improve efficiency and control costs.

First, we worked on correctly assigning trucks to warehouses (distributors), using the initial number of trucks registered at each warehouse as a basis and calculating their proportion of the total fleet. Using this proportion, we dynamically redistributed the fleet based on the total number of trucks defined, ensuring that the allocation reflected current demand and avoided over or underutilization of resources.

The core of the model was implemented in a statechart within the Truck agent, which defines the states and transitions a truck goes through during a delivery trip. The cycle begins in the `atDistributor` state, where the truck picks up a pending order,

and then moves to the corresponding retailer (movingToRetailer). Upon arrival, delivery and throughput times are recorded, unloading is simulated with a probabilistic wait time (unloading), and then a check is made to see if it can continue operating within business hours (CheckingWorkingSchedule). If the truck can continue, it returns to the warehouse (MovingToDistributor), closing the tour and allowing a new one to be started if there are pending orders.

To calculate distances, many variables were created and used, which determines the distance between trucks, warehouses, and retailers, optimizing the route by always selecting the closest destination with pending orders. This reduces the total distance traveled and associated variable costs, such as fuel consumption and wear and tear.

Operating constraints and rules were integrated, such as maximum truck capacity (5,000 liters), compliance with working hours, and the possibility of a truck continuing its route if it is in transit at the end of the shift, but must pause and resume the tour on the next business day. In addition, loading and unloading times were considered with triangular distributions to reflect the realistic variability of these processes.

Finally for the truck programming part, the daily distance traveled by each truck was tracked using a cumulative variable, which facilitates the calculation of variable costs and performance analysis.

Once we finished adjusting the relationships between our agents and configured the scenarios to run simulations as mentioned in the problem, we focused exclusively on collecting two of our most important KPIs throughout this case study: totalcostaccumulated() and totalspoilageaccumulated(). The first of these was the collection and summation of observed costs across all our agents, such as the fixed costs of ordering new material from distributors to the variable costs of our fleet depending on the number of kilometers traveled by the trucks. While it seems like a fairly simple task, collecting data across multiple agents forced us to delete some graphs that didn't work correctly, but we relocated them to their corresponding agent to avoid problems and errors in the model.

To conclude that our model was working correctly we made sure of the next metrics:

- The events were triggering on time and correctly.
- Our cost was variable and fixed during time because of the different types of variables.
- Our fleet was moving realistically.
- The distribution of our graphs made sense and are realistic for a company like LactoTrans.

The key logic of our model lie in a work schedule where multiple actions are performed on Saturdays so as not to penalize the lack of work on Sundays. Similarly, we have a schedule for our truck fleet, and its movement is based on a `statechart()` programmed with the needs of the problem. Both our distributors and retailers have independent functions created in the same agent to map metrics of interest to us. Each of these metrics fulfills a function that is ultimately weighted with more general functions in main such as the total accumulated cost or the number of delayed orders. In summary, the strength of our model lies in its functions and the ability to trigger said functions over events that are realistic and meet what the model requires. Each function and event are independent but is combined at the end of the days or weeks depending on the metric being analyzed.

Data

To support the model development, we created and updated the distributor and retailer tables to serve as the core database for our simulation. These tables were enriched with the necessary attributes such as location coordinates, capacity levels, and truck allocations, ensuring they could drive the logic of order assignment, routing, and inventory management effectively. The final structure and content of these tables can be seen in Figures 6 and 7 in the Appendix.

Assumptions and Their Validity

One important simplification was regarding the truck dispatch process. Although the real world logistics might involve dynamic routing, varying loading times, and potential breakdowns, the model assumes a simplified delivery logic where trucks start from their warehouses, deliver orders following a nearest neighbor route, and return to base. This helps focus on capacity and inventory constraints without getting bogged down in overly detailed route optimization.

Another key assumption relates to the refill costs at distributors. Following the problem statement, the model does not include explicit refill costs for distributors who already have stock. However, in experiments where we allowed increased demand at two warehouses randomly, we observed in “*figure 3*” on the appendix a significant increase in spoilage due to excess inventory sitting too long. This suggests that ignoring refill costs can lead to overstocking and product wastage, highlighting a potential area for future model refinement.

Because the problem did not specify which warehouses to expand in the capacity increase scenario, we applied random selection of one or two warehouses for expansion. This allowed us to test the effects of capacity increase broadly, though in practice, expansion decisions would likely be data-driven or strategic.

Additionally, we assumed that order arrivals are independent and identically distributed across retail partners, ignoring possible seasonal or promotional demand spikes. While this simplifies demand modeling, it may overlook important temporal demand patterns that could affect inventory and delivery performance.

Finally, the model assumes constant travel times and traffic conditions based on static GIS data, without incorporating potential delays due to congestion or weather. This assumption is typical for strategic level simulations but may not capture operational variability experienced daily.

Overall, these assumptions strike a balance between realism and model tractability. They are valid within the scope of this study but point to areas for further data collection and model enhancement as the project evolves.

Model Outputs (KPIs)

The primary Key Performance Indicators (KPIs) for the simulation focus on service level and operational efficiency. The most critical measure of customer service is the probability of on-time delivery to retail partners, defined as deliveries made by the next working day after order placement. Orders not delivered within this timeframe are classified as delayed, primarily caused by either inventory stock-outs at warehouses or insufficient truck capacity.

Other important KPIs include:

- Total costs, encompassing transportation, storage, spoilage, and fixed costs per order. We can see this metric in figure 2 in the appendix.
- Order delay probabilities by retail partner, reflecting how frequently customers experience late deliveries.
- Spoilage rates, particularly relevant for perishable products such as milk, yogurt, and cheese.
- Distance traveled per truck to evaluate variable transportation costs and route efficiency.
- Fleet capacity utilization, including comparisons when introducing larger trucks with higher fixed and variable costs.

These KPIs collectively provide a comprehensive view of LactoTrans's distribution network performance from financial, operational, and customer satisfaction perspectives.

Scenarios and Experimentation

Baseline Analysis:

The current network configuration was modeled over a 10-15 week horizon as seen in “figure 5” on the appendix, representing typical operational variability. This established baseline values for total cost, delay probabilities, stock-outs, and fleet utilization. Results showed significant challenges with late deliveries and inventory shortages, reflecting an undersized truck fleet and inefficiencies in order scheduling and warehouse capacity.

New Warehouse Scenario:

A new warehouse in Târgu Mureș was introduced to assess its potential impact. Retail partners were reassigned to optimize service coverage and minimize transportation distance. Despite some improvement in local delivery times, the simulation revealed that opening this new warehouse substantially increased overall fixed and variable costs, mainly due to underutilization of trucks and overlapping service areas. Random demand increases at two warehouses during simulations

further emphasized that without expanding the retailer base or improving logistics, cost inefficiencies remain high.

Warehouse Expansion Scenario:

Capacity expansions of 40% were tested for one or two existing warehouses, followed by reallocation of retail partners to balance loads. This approach reduced stock-outs and delays more effectively than the new warehouse scenario by leveraging existing infrastructure. However, higher capacity alone did not fully resolve delivery delays due to insufficient truck fleet size.

Fleet Size and Composition Analysis:

Increasing the fleet improved service levels significantly by reducing delayed orders and enabling multiple daily delivery tours per truck. However, the higher fixed costs of larger trucks necessitate careful cost benefit analysis. The recommendation is to increase the truck fleet proportionally with demand and warehouse capacity, potentially blending standard and larger trucks to optimize both capacity and costs.

Ordering and Inventory Policy Experiments:

Dynamic ordering policies were explored by adjusting order frequency and size based on real-time inventory levels and demand forecasts. Unlike the current practice of immediate full capacity replenishment after each order, separating order placement from delivery receipt and introducing fixed receipt windows enhanced flexibility and reduced spoilage. Using Monte Carlo simulations to vary demand and supply lead times, policies prioritizing smaller, more frequent orders showed better performance in reducing stock-outs and overstocking, thus optimizing cash flow and minimizing waste.

Results and Analysis

Firstly, linking order placement directly to immediate delivery receipt causes inflexibility, leading to inventory overstocking or stock outs, increased spoilage, and higher holding costs. Separating these processes allowed for strategic delivery scheduling that respects warehouse capacity and current demand.

Secondly, the new warehouse in Târgu Mureș, while intended to improve service coverage, resulted in higher total costs without proportional service level gains. This was exacerbated when simulated demand randomly increased at certain warehouses, showing that expanding capacity without scaling the retailer base or truck fleet leads to inefficiencies. It also turns the warehouse Iasi obsolete as we see in “figure 1” in the appendix.

Thirdly, the truck fleet size is a limiting factor in timely order fulfillment. Insufficient trucks lead to accumulated delayed orders, harming customer satisfaction. Increasing fleet size and improving route planning through a nearest-neighbor approach reduced travel distances and delivery times, enhancing overall efficiency.

Fourthly, dynamic inventory management based on up to date stock levels greatly improves responsiveness and reduces waste compared to static replenishment parameters.

Lastly, fixed and variable costs must be tracked closely. Larger trucks offer capacity benefits but come with increased fixed and per-kilometer costs that must be balanced against service improvements.

Conclusion and Future Work

In conclusion, the simulation model provides valuable insights into LactoTrans’s distribution challenges and opportunities. The main recommendations are:

1. Avoid opening the new warehouse unless retailer coverage is significantly increased to justify the added costs.
2. Prioritize expanding the capacity of existing warehouses and correspondingly increasing truck fleet size.
3. Separate order placement and delivery receipt processes to enhance operational flexibility.
4. Implement dynamic, data-driven inventory replenishment policies to reduce spoilage and stock-outs.
5. Employ route optimization based on geographic proximity to minimize distance traveled and related costs.

6. Monitor truck utilization and travel distances continuously to inform fleet management decisions.
7. Consider a mixed fleet of standard and larger trucks, balancing cost and capacity based on demand patterns.

Future work should include integrating real-time data feeds for demand and inventory, exploring stochastic variations in supply chain disruptions, and testing advanced routing algorithms such as vehicle routing problem (VRP) solvers. Additionally, incorporating environmental impact metrics and driver scheduling constraints would further refine the model's applicability.

Appendix:

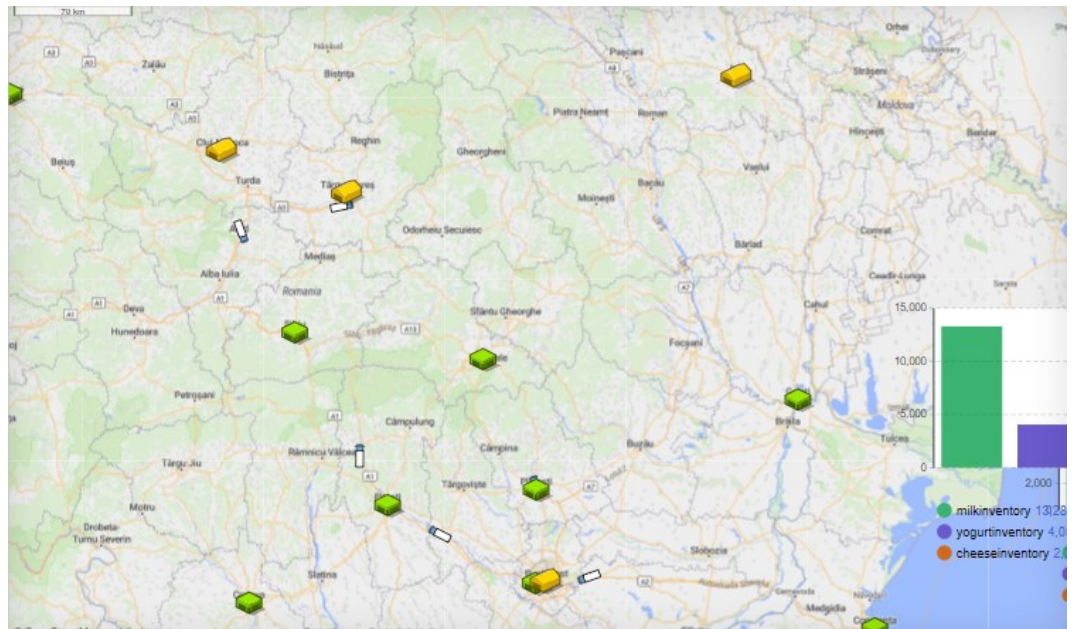


Figure 1: Iasi becomes obsolete warehouse

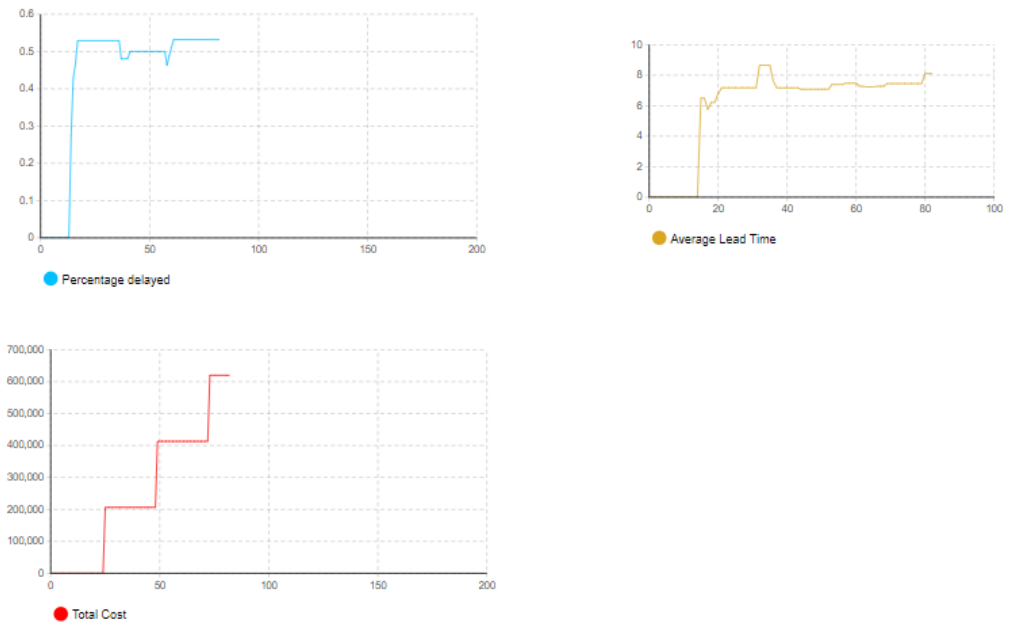


Figure 2: Total cost and delays increasing

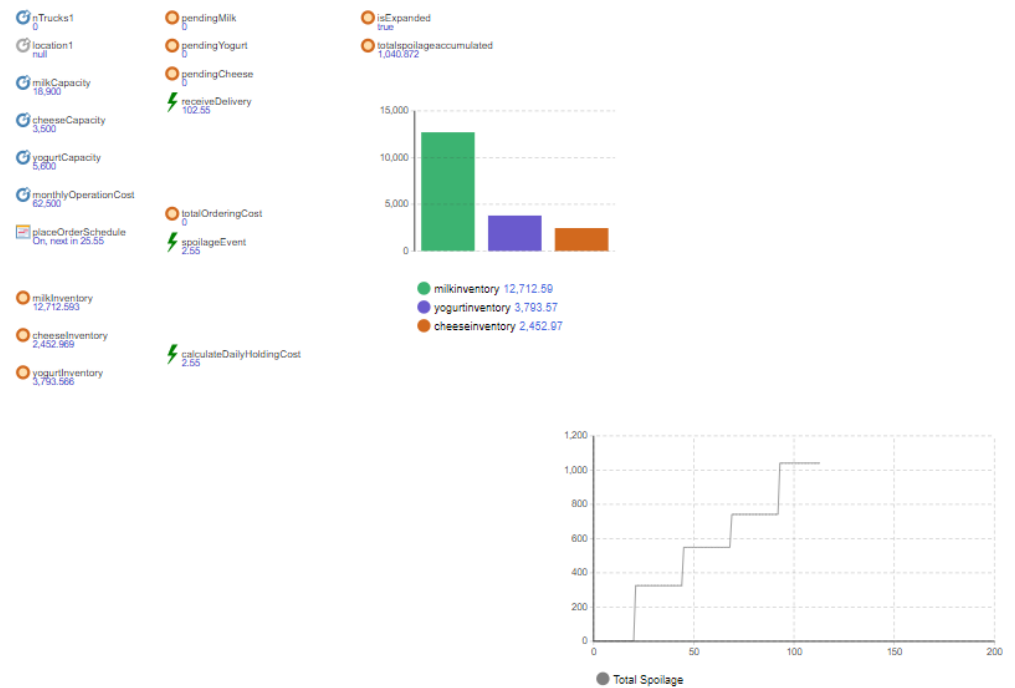


Figure 3: Spoilage Increasing thanks to fulfilling maximum capacity



Figure 4: Variables for Trucks

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Initial number of trucks:8
Percentage of trucks in Bucharest is 0.25
Percentage of trucks in Cluj--Napoca is 0.25
Percentage of trucks in Iasi is 0.25
Percentage of trucks in Târgu Mureş is 0.25
Bucharest ready to place orders.
Cluj--Napoca ready to place orders.
Iasi ready to place orders.
Târgu Mureş ready to place orders.
Iasi expanded (random selection)
Bucharest expanded (random selection)
Retailer Craiova ejecutando generateDailyDemand en tiempo 19.0
milkMean=480.0, yogurtMean=144.0, cheeseMean=96.0
Demanda generada hoy: Milk=1005.1718626344887, Yogurt=239.84214744886796, Cheese=40.150596220293934
Retailer Craiova places order at day 0: Milk=1005.1718626344887, Yogurt=239.84214744886796, Cheese=40.150596220293934
Retailer Galaţi ejecutando generateDailyDemand en tiempo 19.0
milkMean=440.0, yogurtMean=132.0, cheeseMean=88.0
Demanda generada hoy: Milk=362.6733111104113, Yogurt=186.32853749331937, Cheese=94.81285721513692
Retailer Galaţi places order at day 0: Milk=362.6733111104113, Yogurt=186.32853749331937, Cheese=94.81285721513692
Retailer Braşov ejecutando generateDailyDemand en tiempo 19.0
milkMean=400.0, yogurtMean=120.0, cheeseMean=80.0
Demanda generada hoy: Milk=42.62671859763553, Yogurt=81.73724599703763, Cheese=81.60250717956713
Retailer Braşov places order at day 0: Milk=42.62671859763553, Yogurt=81.73724599703763, Cheese=81.60250717956713
Retailer Sibiu ejecutando generateDailyDemand en tiempo 19.0
milkMean=280.0, yogurtMean=84.0, cheeseMean=56.0
Demanda generada hoy: Milk=317.02107102260436, Yogurt=72.21293442552351, Cheese=16.725618701877863
Retailer Sibiu places order at day 0: Milk=317.02107102260436, Yogurt=72.21293442552351, Cheese=16.725618701877863

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Figure 5: Correct Calculations of our model

location	n_trucks	milkcapacity	yogurtcapacity	cheesecapacity	monthlyoperatingcost	fixedordercostmilk	fixedordercostyogurt	fixedordercostcheese
Bucharest	2	13500	4000	2500	50000	120	80	150
Cluj-Napoca	2	13500	4100	2800	45000	120	80	150
Iaşi	2	13500	4400	2900	47000	120	80	150
Târgu Mureş	2	11000	3500	2100	40000	120	80	150

Figure 6: Distributor Final Data Table

location	assigneddistributor	yogurtmean	milkmean	cheesemean
Ploieşti	Bucharest	120	400	80
Constanţa	Bucharest	144	480	96
Piteşti	Bucharest	96	320	64
Timişoara	Cluj-Napoca	168	560	112
Arad	Cluj-Napoca	72	240	48
Oradea	Cluj-Napoca	108	360	72
Braşov	Iaşi	84	280	56
Galaţi	Iaşi	120	400	80
Craiova	Iaşi	132	440	88
Sibiu	Iaşi	144	480	96

Figure 7: Retailer Final Data Table

PART C: Critical essay of two simulation papers from 2024

In this essay we are going to examine two simulation case studies published in 2024, each employing a different simulation technique to solve real-world problems. Simulation is a great tool used across industries to model, test, and optimize complex systems (Law, 2014). The first paper uses Monte Carlo Simulation (MCS) to analyze the risk of fluoride exposure in Iran. While the second paper applies Agent-Based Modeling (ABM) to study urban mobility and energy consumption patterns. The goal of this essay is to discuss the choice of technique used by the company, and to discuss whether an alternative method could have been applied instead of the chosen approach.

Paper 1: Evaluation of fluoride exposure using disability-adjusted life years (DALYs) and probabilistic risk assessment

This paper is trying to manage a public health problem which is people being exposed to too much fluoride, which can cause dangerous health issues like fluorosis (teeth and bone problems). The researchers wanted to understand how harmful this exposure could be for the population over time. The challenge was that fluoride levels and how people react to them can vary a lot. This means there is uncertainty in the data because not everyone drinks the same water, eats the same food, or has the same health in general. Because of this uncertainty and variation, the authors chose to use Monte Carlo Simulation. MCS is a technique that can simulate thousands of possible scenarios to show how different levels of fluoride exposure could affect people. Monte Carlo simulation is often used when a system has uncertainty and variability and helps give a range of possible outcomes, rather than just one fixed answer. (Law, 2014)

For this specific case Monte Carlo is most suitable, but we will evaluate if DES (MCS) or Agent-Based Modeling (ABM) could be alternative approaches. Using a different technique like Discrete-Event Simulation or Agent-Based Modeling would have been less suitable for this problem. This because DES is good for modeling systems with clear steps and events over time flows. But in this case, the problem is not about a process or system with clear events. It's about estimating health risks from exposure of fluoride, which is not event driven. Therefore, DES wouldn't be a good fit. ABM models the behavior of individual agents and how they interact in an environment. It could maybe be used here to model individual people's behavior like how much water they drink or other factors that contain fluoride, but it would be much harder to simulate compared with using MCS. Since the main goal is to understand overall health risk for a population, not how individuals behave, ABM is not the best option but could be possible to use if they focus on different age groups or more specific groups. Monte Carlo Simulation was the best choice because it's great at dealing with uncertainty, probability, and population-level risk (Law, 2014). Discrete-event simulation is best for systems with queues and events, (Banks et al, 2010) while agent-based modeling is good when individual behaviors and interactions patterns matter (Macal, 2010).

Paper 2: Agent-Based Modeling for Urban Mobility and Energy Use

This paper presents an agent-based model (ABM) and looks at urban mobility and energy consumption, where people's travel behavior plays a big role for the life and environment they live in. The goal of the study is to understand how different energy policies might affect people's travel decisions and energy use in cities. As well as optimizing and designing infrastructure for a better and more efficient daily life for the inhabitants. The challenge is that people behave differently, some prefer public transport, some use private cars, and their habits can change depending on income, location, or policy changes. Because of this variation and the need to model individual decisions the authors chose to use Agent-Based Modeling (ABM). ABM was chosen because the study aimed to model the actions and interactions of autonomous agents like consumers, transit operators and households with distinct behaviors and decision rules. ABM allows the researchers to simulate each person as an agent with specific rules and behaviors. It's useful in this case because it enables the quantification of the impacts of various urban planning decisions on a given scenario.

For this specific case, ABM is most suitable, but we will evaluate if Monte Carlo Simulation (MCS) or Discrete-Event Simulation (DES) could be alternative approaches. Using Monte Carlo Simulation would have been less appropriate here because MCS is best for problems involving random variation and uncertainty in inputs, where the system does not depend on interactions between individuals. MCS could maybe estimate energy demand under uncertainty, but it cannot model how people's behavior and choices change over time based on the system and environment around them. DES would also be less suitable because it models structured processes and queuing systems with clear events and steps. The system in this paper is not about queues or processes, but it's about human decision-making and interactions over time. Therefore, ABM was the best choice, as it can model individual travel behavior and show how different energy policies might affect people differently depending on their characteristics and choices (Macal, 2010). While other methods can model some parts of the system, only ABM gives the detail needed to understand behavior-driven policy outcomes.

Conclusion

These two case studies highlight the importance of choosing a simulation technique that aligns with the specific features of a problem and what the main goal for the simulation is. Monte Carlo simulation and Agent-Based Modeling each have strengths suited to different types of systems: Monte Carlo for quantifying uncertainty and evaluating outcomes under probabilistic variation, and ABM for capturing behaviorally complex, dynamic interactions between agents. Through careful problem definition and technique selection, simulation can offer deep insights into system behavior, generate good decisions, and reduce real-world uncertainty.

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